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Reuse-Aware Routing for Energy-Efficient AI Inference on Heterogeneous GPUs

Research Proposal submitted to the International Academic Research Competition (IARCO) 2026

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1 INTRODUCTION

Growing concerns over the energy and water demands of data centres have prompted authorities in several countries to impose or consider moratoria and other development restrictions amid the expansion of artificial intelligence [1]. Despite these debates, AI has experienced rapid growth in fields ranging from healthcare to finance, such as fraud or cancer detection recognition [2]. Agentic-AI in particular, an autonomous system executing multi-step workflows, has been adopted and scaled by roughly 40% of large organizations [3]. Given such circumstances, it is impractical to simply limit the use of AI; we must instead address the challenge of balancing its advantages with its environmental footprint: a pressing dilemma that must be resolved to ensure sustainable development.

A growing proportion of this environmental cost comes from AI inference, which is the process of continuously running a deployed AI model. As request volume increases, inference computation becomes an increasingly important source of energy consumption [4]; the International Energy Agency's 2025 report estimates that electricity use by data centres worldwide produces approximately 180 million tonnes of CO_2 annually [5]. The efficiency of AI inference partially depends on hardware: Graphic Processing Units (GPUs) exhibit heterogeneous characteristics and may consume different amounts of energy [6].

Nevertheless, workloads presented to the GPU is determined not only by the request itself. Key-Value (KV) caching, also known as Prefix caching, is an optimization strategy that stores Key and Value vector computations in memory, reducing repeat prefill computation [7]. As previously mentioned, agentic-AI has become increasingly popular but still remains a largely unexplored area in academic research, introducing multi-turn, non-linear execution, and an unusually high observed prefix reuse [8]. Additionally, different GPU architectures experience different levels of energy reduction [6]. Hence, a router that ignores prefix reuse may select a suboptimal GPU for a request's effective workload. This study proposes to characterize the relationship between prefix-reuse ratio and inference energy across heterogeneous GPUs and develop a router that selects the lowest-energy GPU while maintaining latency requirements.

2 RESEARCH QUESTION

How can we fully leverage the high KV-cache reuse opportunities inherent in agentic AI serving systems and exploit the different energy characteristics of heterogeneous GPUs to design a system that balances sustainability with performance?

3 LITERATURE REVIEW

3.1 Heterogeneous GPUs and inference energy

Previous studies have explored how hardware architecture may impact the energy consumed during inference. Argericha et al. ran over 5,000 energy-and-performance experiments on 50 LLMs and 10 NVIDIA GPU types on offline batch inference and online server inference under various load conditions, building up to a benchmark and dataset called Watt Counts [6]. In offline batch inference, the H100 achieved the lowest energy per token for 90% of the models, but the L4 GPU was most efficient for three models. Furthermore, using H100 for extra-large models consumed approximately 20% less energy despite the H200 achieving higher throughput [6]. Collectively, these results demonstrate that the energy-optimal GPU depends on interactions

between model architecture, GPU architecture and deployment scenario rather than hardware performance, providing an empirical foundation for heterogeneous energy-aware routing.

He et al. shows how the observation from Watt Counts can be operationalized through Fair, Resource- and Energy-Efficient Scheduling (FREESH), a system that selects GPU types, tensor-parallel configurations, regions, worker counts and frequencies according to request length, performance, energy and carbon characteristics [9]. Compared with its baseline without a request classifier, FREESH reports a 28.60% reduction in energy and a 45.45% reduction in operational emissions over the one-hour experiment [9]. Similarly, Stojkovic et al. shows that inference energy depends on serving configuration, dynamically adjusting model-instance count, tensor parallelism and GPU frequency according to workload demands. However, this study focuses on configurations within a H100-based setting rather than across GPU architectures [10].

3.2 KV Caching and sustainability

Existing studies have demonstrated that managing KV-cache properly can largely reduce the carbon emissions in serving systems. GreenCache [11] improves sustainability by dynamically adjusting the KV-cache size during runtime. It characterizes tradeoffs between the operation carbon (greenhouse gas emitted from electricity used to power servers) and extra embodied carbon (emissions spent making the hardware components for inference), successfully reducing carbon emissions by up to 25.3% [11]. Additionally, “Store or Recompute” presents an analysis on the carbon tradeoff between storing KV cache or recomputing. They found out that the optimal storage time is three minutes before the carbon cost of storing the KV cache exceeds the carbon cost of recomputing it [12]. Wang et al. conducted the first large-scale characterization of cache-reuse patterns [7], identifying reuse opportunities and proposing a new cache eviction policy to improve the cache-hit rate, thereby minimizing the operational carbon described in GreenCache.

3.3 Gap in the Literature

Although Watt Counts [6] and FREESH [9] exploit the relationship between GPU types and energy consumption, these studies largely ignore the opportunities and influence of reusable prefix caching in agentic systems. Additionally, GreenCache [11] and Wang et al. [7] identify the importance of KV cache management for saving energy costs but fail to leverage the differences between GPUs and their energy consumption behaviours, which overlooks a significant optimization opportunity in existing heterogeneous GPU clusters.

4 METHODOLOGY

4.1 Research Design

This study will use a quantitative experimental approach through controlled GPU measurements and a simulation created under such conditions. The research will proceed in two stages. First, it will characterize how increasing KV-prefix reuse affects inference energy and latency across heterogeneous GPUs. This will establish whether the resulting reduction in prefill computation changes which GPU is most energy-efficient for the same request. Second, the study will evaluate the carbon and SLO impact of incorporating these measured relationships, together with time-varying carbon-intensity signals, into request routing and KV-cache management.

GPU heterogeneity will be characterized through measured differences in energy consumption and execution time across workloads and KV-reuse levels. Sustainability will be evaluated in terms of operational carbon emissions, calculated from measured energy use and the carbon intensity of electricity at the time and location of each operation. The primary study will use one fixed model and two GPU types (RTX 3090 and H800) with contrasting performance and power characteristics. Since installed hardware capacity remains fixed throughout the study, embodied carbon is outside the initial accounting boundary. Accordingly, the findings will concern serving-related operational emissions rather than the complete hardware lifecycle. Measurements from this controlled testbed will then calibrate a simulation that extends the evaluation to a broader range of workload and carbon-intensity conditions.

4.2 System Architecture

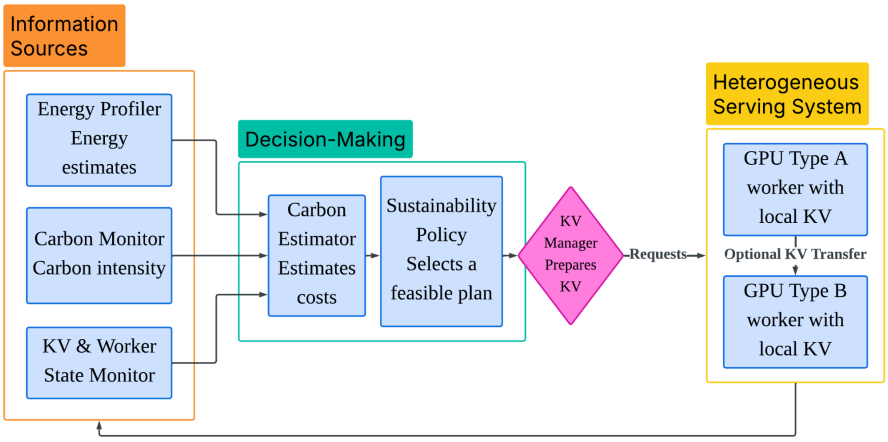


Figure 1: *Implementation pipeline, consisting of information sources, decision making, and heterogeneous serving*
Source: by the author

The energy profiler in figure one above characterizes execution and KV-movement energy, while the monitors provide carbon intensity alongside information on compatible cached prefixes and worker queues. For each arriving request, the estimator predicts execution-plan carbon costs, and the policy selects optimal execution plan subject to latency and memory constraints. The KV Manager will then prepare any required cached state and dispatch the request to the selected worker. Runtime energy, latency, and resulting KV locations will be recorded and fed back into the system state. The system will be implemented on NVIDIA Dynamo, which supports KV-aware routing, with vLLM as the inference backend [13]. We will extend the routing logic to evaluate both direct and assisted plans described below.

Energy profiling: Following the empirical profiling approach used in Watt Counts [6], the study will measure hardware behaviour by controlling model weights, numerical precision and serving settings, while varying input length, output length, batch size and reusable-prefix fraction. Reuse fractions of 0%, 20%, 40%, 60%, 80% and 95% will be created by pre-populating the corresponding prefix KV while leaving the input request unchanged. The actual number of reused tokens will be recorded. Measurements will distinguish between prefill, decode and KV-transfer, as avoiding computation and moving cached state impose different costs.

A Python experiment harness will collect timestamped execution logs and GPU power readings through the NVIDIA Management Library (NVML). After warmup, each profiling configuration will be measured five times, restoring the prescribed cache state before each run. CPU, memory, network and controller costs will be measured where possible or explicitly estimated, avoiding overlap between device and whole-node measurements.

Operational-carbon model: The profiler will store energy measurements, which the Carbon Estimator will combine with the applicable carbon signal. For request r, worker w, reusable-prefix fraction h and execution time t, we define estimated execution carbon as:

$$\widehat{C}_{exec}(r,w,h,t) = \frac{\widehat{E}_{exec}(r,w,h,t)}{3.6 \times 10^{6}} CI_{z(w)}(t)$$

Source: GreenCache [11]

Here, energy is measured in joules and $z(w)$ identifies the worker's location. Carbon intensity CI is measured in gCO_2e/kWh and the resulting carbon cost gCO_2 [14]. Movement carbon applies the carbon intensity relevant to transmitting and installing the KV state on each component. Longer operations will be divided across carbon-signal intervals.

Request workflow: When a request arrives, the state monitor will identify the compatible prefix already available on each worker. The policy will then construct a direct plan using local KV and, where supported, an assisted plan that imports additional compatible KV. Their costs will be compared as follows:

$$\widehat{C}_{direct} = \widehat{C}_{exec}(h)$$

$$\widehat{C}_{assist} = \widehat{C}_{move}(\Delta K) + \widehat{C}_{exec}(h')$$

Source: By author

The request, worker and time arguments are omitted here for readability. The term ΔK represents the additional KV required, and h' is the resulting larger reusable prefix. Hence, the assisted plan accounts for both the computation avoided and the movement required to obtain that benefit.

1 Plans will be screened against the specified time-to-first-token (TTFT) and time-per-output-token (TPOT) limits, including relevant queuing and transfer delays, and against memory requirements including output growth. Unknown output length will be estimated from prior requests. The policy will choose the eligible plan with the lowest estimated carbon cost. When a feasible direct alternative exists, a transfer must improve upon the best direct plan across all workers by a margin that accounts for estimation error. If only an assisted plan is feasible, it may be selected; if no plan is feasible, a predefined admission rule will apply consistently across experiments. The KV Manager will reserve resources, perform any transfer and dispatch the request. Actual latency, energy and resulting prefix locations will then be recorded.

Workloads: The Qwen-Bailian To-C and Coder traces will be used alongside controlled shared-prefix workloads, preserving key workload characteristics [7]. Recorded output lengths will drive replay, while routing uses predicted lengths. Historical Great Britain carbon-intensity

data from NESO [14] and a constant-intensity control will represent labelled simulation scenarios. Larger-scale experiments will use Dynamo Mocker with measured GPU timing profiles and an added energy model based on execution and KV-transfer measurements.

The proposed policy will be compared with Dynamo's KV-aware router [13], an energy-aware policy that ignores worker-specific cached prefixes, and a local-KV-only variant that disables assisted plans. Prefix caching will remain enabled for all policies. Comparisons will use identical workloads, worker resources, cache capacities and eviction rules, initial cache states, carbon signals and evaluation periods. Admission rules, TTFT/TPOT limits and replay settings will be fixed across policies.

5 EXPECTED RESULTS

Increasing KV-prefix reuse is expected to reduce prefill energy and latency, although the magnitude may differ across workloads and GPU types. These differences may change the lowest-energy feasible worker for certain request configurations. A GPU-ranking crossover is not assumed; unchanged rankings will also help establish the limits of reuse-aware hardware selection.

At the system level, an initial experimental target is 10–20% lower total serving energy than the prefix-unaware variant on prefix-rich workloads with uneven cache placement and multiple feasible destinations. For context, FREESH reports 28.6% energy savings [9], while GreenCache reports an average 15.1% reduction in carbon emissions, reaching 25.3% [11].

Assisted plans are expected to provide additional savings when avoided execution costs exceed movement costs, although no fixed improvement over the local-KV-only variant is assumed. Benefits may approach zero or become negative when reuse is limited, transfers are expensive or queuing restricts routing choices. Under constant carbon intensity, percentage operational-carbon savings will equal percentage energy savings; time-varying scenarios will be evaluated separately. The target is to achieve these savings without reducing the proportion of submitted requests completed within both latency limits. Overall, the findings will identify when reuse-aware hardware selection and KV movement improve serving efficiency and when their costs outweigh their benefits.

6 CONCLUSION

This study addresses the challenge of reducing the environmental footprint of agentic AI inference while maintaining performance. It investigates how KV-prefix reuse and the different energy characteristics of heterogeneous GPUs can be jointly exploited through reuse-aware routing. By measuring execution and KV-transfer costs, the study will develop and evaluate a policy that selects workers and cache-transfer plans under latency and memory constraints. The findings are expected to establish when this approach reduces energy consumption and operational emissions without compromising service, and when movement costs outweigh its benefits. This would provide practical evidence for supporting growing AI demand through more efficient use of existing computing resources.

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